Ashvin Gupta

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PhD researcher at Imperial College London on the AI4Health programme, developing machine-learning methods for longitudinal electronic health records. Research interests include representation learning, language diffusion models, mechanistic interpretability, and world models, with previous industry experience in computer vision and medical imaging.

Education

Imperial College London **2024–2027**
PhD in AI and Machine Learning

Imperial College London **2019–2023**
Biomedical Engineering, First Class Honours, Dean's List Years 1–3

Hymers College **2009–2019**
A-Levels, 4 A*s, Head Boy

Research Experience

Imperial College London **2024–Present**
PhD Researcher, AI and Machine Learning

- Developed a two-stage framework that textualises structured clinical events and adapts pretrained causal LLMs through LoRA continued pretraining and discriminative fine-tuning. Using Mistral-7B, achieved the strongest performance across four MIMIC-IV outcomes and pancreatic-cancer prediction in CPRD, outperforming the best baselines by 0.013–0.021 AUROC on MIMIC-IV.
- Investigated block-based masked diffusion as a non-autoregressive alternative for zero-shot clinical outcome prediction from longitudinal EHRs. Developed capacity-matched diffusion and autoregressive models on MIMIC-IV, showing up to 1.43× faster inference while retaining comparable learned clinical representations.
- Adapted JEPA-style self-supervised learning to longitudinal EHRs using three different masking strategies. Evaluated representations across four MIMIC-IV clinical prediction tasks and identified key limitations of JEPA formulations for EHR modelling.
- Developed an end-to-end in-context learning pipeline to translate NICE pancreatic and lung cancer guidelines into executable Answer Set Programs, producing interpretable patient-specific recommendations.
- Developing a Transformer-based model with QMUL for early pancreatic-cancer prediction from 16 million CPRD primary-care records, modelling patient histories across clinically relevant prediction horizons.
- Investigating representations learned by the "EveryQuery" EHR foundation model using linear probes, sparse autoencoders (SAEs), and J-Lens to identify clinically meaningful features and understand how patient information is encoded across model layers.
- Developing an EHR world model that explicitly separates clinical observations from actions/interventions and learns predictive latent-state dynamics using a LeWorldModel-inspired joint-embedding objective.

Industry Experience

Oxford Heartbeat **2023–2024**
Machine Learning Engineer

- Developed a supervised deep-learning pipeline for cerebral vessel segmentation from DICOM CT scans, fine-tuning a ResNet18-based model to automate vessel thresholding.
- Applied nnU-Net to 3D CT segmentation to identify regions with high probability of haemorrhage, supporting downstream vessel centreline extraction.
- Developed a clinician-facing annotation tool for cerebral CT scans using image-processing methods, reducing vessel-labelling time from several hours to minutes. Presented the work to Innovate UK, contributing to the award of additional project funding.

Bank of America **Summer 2022**
Software Engineer Summer Intern

- Developed a CI/CD pipeline for nightly application stress testing to support risk mitigation. Worked within an Agile development environment and used Git for collaboration on a large-scale codebase.

Publications

1. **A. Gupta**, B. Delaney, A. Russo. "Evaluating Pretrained Language Semantics for Structured EHR Prediction Tasks." *Proceedings of ML4H*, 2026.
2. **A. Gupta**, J. Placidi, B. Delaney, A. Russo. "Block Diffusion for Longitudinal EHR: Accuracy, Efficiency, and Representation Trade-offs." *NeurIPS Workshop on World Models for High-Stakes Health*, 2026.
3. **A. Gupta**, B. Delaney, A. Russo. "Are Joint-Embedding Predictive Objectives Effective for Longitudinal EHRs?" *NeurIPS Workshop on World Models for High-Stakes Health*, 2026. **Selected for Oral Presentation.**
4. **A. Gupta**, D. Prociuk, A. Russo, B. Delaney. "Automatic Conversion of NICE Guidelines to an Executable Computational Model Using Large Language Models." *Learning Health Systems*, 2026.
5. C. Chiu, A. Roy, S. Cechnicka, **A. Gupta**, A. L. Pinto, C. Galazis, K. Christensen, D. Mandic, M. Varela. "Physics-Informed Neural Networks Can Accurately Model Cardiac Electrophysiology in 3D Geometries and Fibrillatory Conditions." *Statistical Atlases and Computational Modelling of the Heart*, 2024.

Teaching

Imperial College London 2025–2027

- **Deep Learning** (Graduate Teaching Assistant): Supported Master's students with coursework covering CNNs, VAEs, and diffusion models.
- **Mathematics for Machine Learning** (Graduate Teaching Assistant): Supported Master's-level teaching in vector calculus, probability, and statistics.
- **Computing Research Collective** (Research Mentor): Mentored five BEng Computing students in critically evaluating research papers on AI applied to healthcare data, explaining deep learning methods and architectures.

Selected Academic Activities

- **Machine Learning Summer School (MLSS), Columbia University, 2026:** Selected as one of 200 participants from 1,200 applicants for a two-week programme covering reinforcement learning for LLMs, causal learning, and time-series modelling.
- **Peer Reviewer:** Reviewed submissions for Structured Data for Health (*ICML 2026*), Beyond Next-Token Prediction (*NeurIPS 2026)*, and Machine Learning for Healthcare 2026 Conference.

Additional Activities

- **Imperial College Lacrosse** – First X Captain (2021–2022), Vice President (2022–2023): Co-led a 60+ member club and organised the UK's largest mixed lacrosse tournament.
- **Sport:** Represented Yorkshire at U15 cricket and Imperial College First XI; completed an Olympic-distance triathlon and was a BUCS Lacrosse National Trophy runner-up.